Mylan		
	Document Name	AIX Migration from 7.1 to 7.2 Through VIO Repository
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1. PURPOSE

This document provides the steps to migrate the AIX OS version from AIX 7.1 to AIX 7.2.

2. SCOPE

2.1. In scope:

2.1.1 If OS version is going end of support, we will communicate it into Business and proposing OS upgrade to N-1 version release.

2.1.2 We will receive RITM from Application/ Database / Business unit for OS upgrade

2.1.3 If we receive email from Application/ Database / Business unit for OS upgrade, we will ask them to raise a RITM

2.1.4 We will raise a Normal change and based on SOP “MYGLO-SOP-IT-CHM-0001” we will get approval, then will implement it

2.2. Out of scope:

2.2.1 To check Proposed OS version is compatible to the database or the application.

2.2.2 Upgrading Database or application version to meet OS support version.

2.2.3 Business approval to upgrade OS, Application or Database

3. PREREQUISITES

3.1 Operating System AIX 7.2 Base ISO Images. We can download it from URL https://www.ibm.com/servers/eserver/ess/ProtectedServlet.wss#toggle

3.2 We can download TL Update from fix central URL https://www.ibm.com/support/fixcentral

3.3 VIO server should have enough free space in rootvg to create virtual media repository.

3.4 AIX Client LPAR must have 40% free disk space in rootvg

4. PROCEDURE

4.1 Login to the Aix LPAR (in which OS will upgrade) with MID and switch to root. Find out the VIO server, Vhost detail to map virtual optical device by issuing below command.

echo "cvai" | kdb | grep vscsi

```
# echo "cvai" | kdb | grep vscsi
read vscsi_scsi_ptrs OK, ptr = 0xF1000000C0128E90
vscsi0      0x000007 0x0000000000 0x0                0bps104_u0bps104-vio1->vhost2
vscsi1      0x000007 0x0000000000 0x0                0bps104_u0bps104-vio2->vhost2
```

4.2 Create the virtual optical media device with SOP "MYL-UNX-SOP-VOD-001" and map to client AIX LPAR.

4.3 Boot the AIX LPAR in SMS mode, meanwhile login HMC with MID and take console of AIX LPAR.

In Manages Systems, Choose "0BPS202--SN068242P" by Type "2" in navigation key field

Note : 0BPS202--SN068242P (Manged Machine where the LPAR is present)

```
Using username "mXXXXXX".
Keyboard-interactive authentication prompts from server:
| Password:
End of keyboard-interactive prompts from server
Last login: Fri Mar 13 05:29:06 2020 from 6z-m5271811t.myl.com

[?1034hmXXXXXX@u0bhmc001:~> vtmenu


 Retrieving name of managed system(s) . . . 0BPS204--SN067FA1R   0BPS202--SN068242P   0BPS201--SN06823DP   0BPS203-SN06E077P   0BPS101--SN10826CP   0BPS102--SN10999BP

 ------------------------------------------------------------
  Managed Systems:
 ------------------------------------------------------------
   1) 0BPS204--SN067FA1R
   2) 0BPS202--SN068242P
   3) 0BPS201--SN06823DP
   4) 0BPS203-SN06E077P
   5) 0BPS101--SN10826CP
   6) 0BPS102--SN10999BP
   7) 0BPS103--SN10437CP
   8) 0BPS104--SN101435R

 Enter Number of Managed System.   (q to quit): 2
```

In Partitions On Manages System, Choose "0bps101_u0bnim001" by Type "1" in navigation key field.

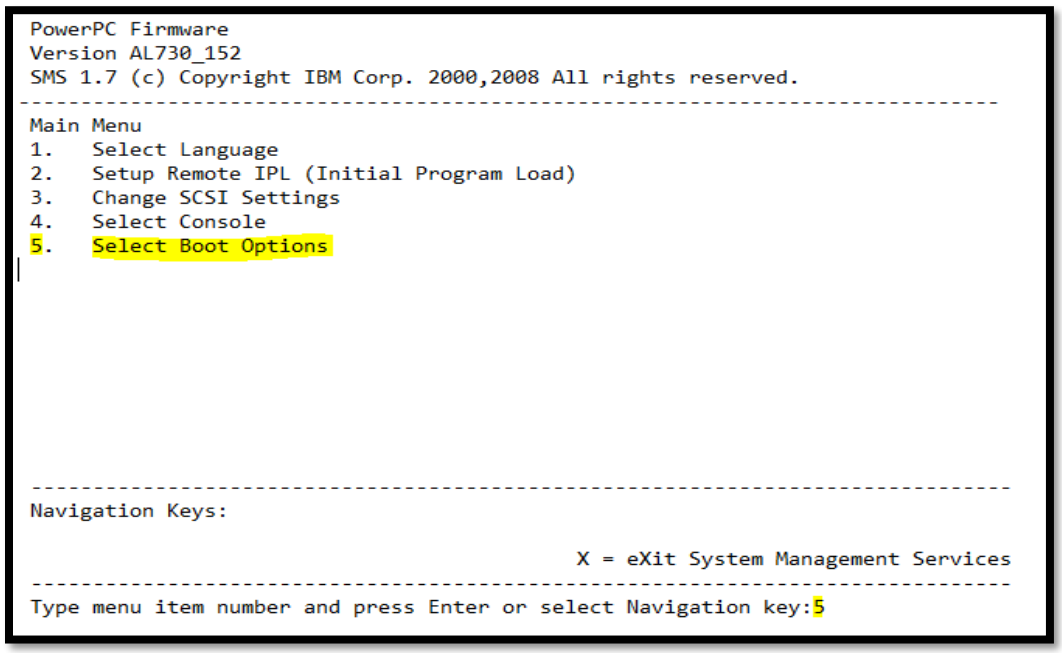
Note : Select LAPR Name which you want to migrate.

```
 ------------------------------------------------------------
  Partitions On Managed System:  0BPS202--SN068242P
  OS/400 Partitions not listed
 ------------------------------------------------------------
   1)    0bps101_u0bnim001                  Running
   2)    0bps103_u01sap120                  Running
   3)    0bps103_u0bdcm007                  Running
   4)    0bps201_u0btvsd004                 Running
   5)    0bps202_maxsp234                   Running
   6)    0bps202_u0bps202-srv               Running
   7)    0bps202_u0bps202-vio1              Running
   8)    0bps202_u0bps202-vio2              Running
   9)    0bps203_bax-sap01                  Running
   10)   0bps203_max-bw06                   Running
   11)   0bps203_max-sap81                  Not Activated
   12)   0bps203_u01sap104                  Not Activated
   13)   0bps203_u0bargust052               Not Activated
   14)   0bps203_u0bsap163                  Running
   15)   0bps203_u0bsap214                  Not Activated
   16)   hax-mlltest                        Running
   17)   obps203_u0bargust051               Running
   18)   u01sap123                         Running
   19)   u0bps303-maxsp201                 Not Activated
   20)   u0bps303-maxsp204                 Not Activated
   21)   u0bps303-maxsp301                 Not Activated
   22)   u0bps303-maxsp304                 Not Activated
   23)   u0bsap181                         Running

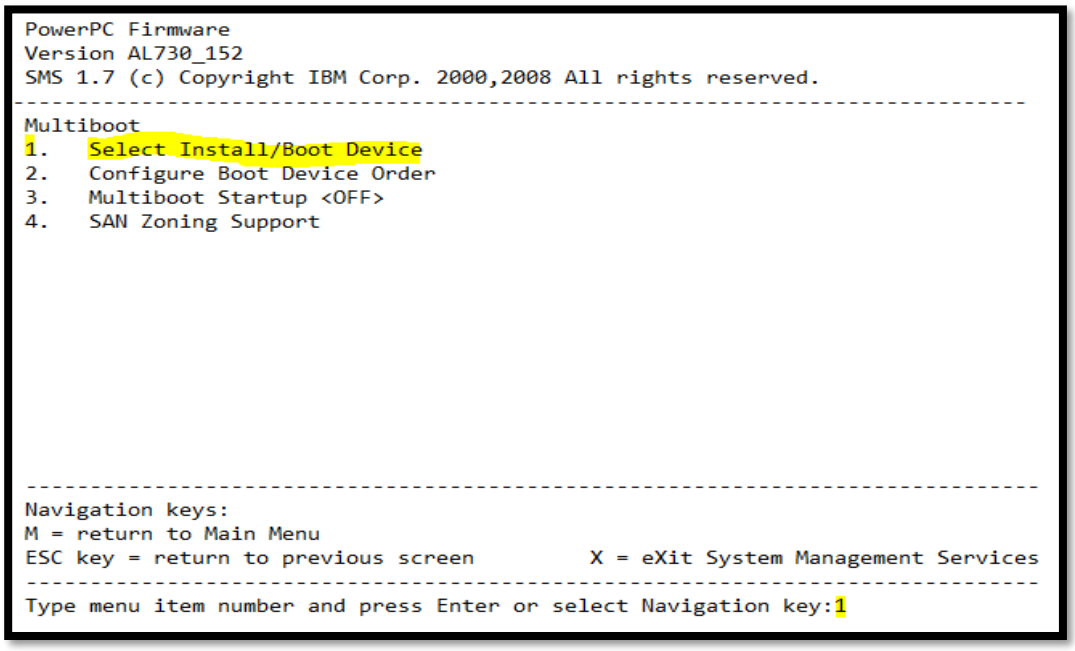

 Enter Number of Running Partition (q to quit): 1

    Opening Virtual Terminal On Partition 0bps101_u0bnim001 . . .
```

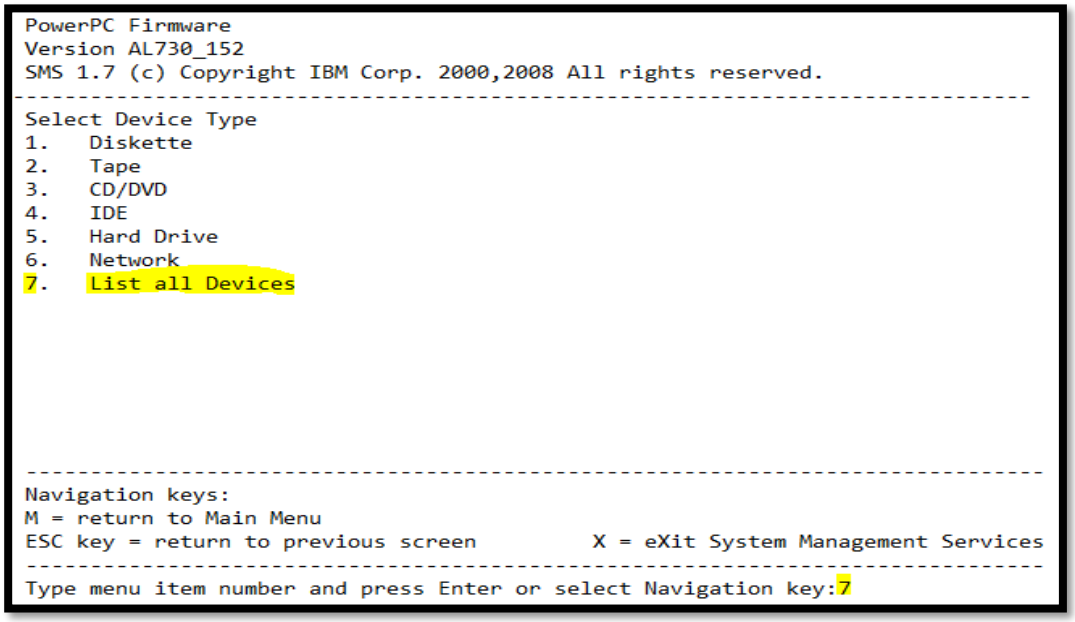
4.4 In SMS mode menu, Choose “Select Boot Options” by Type “5” in navigation key field



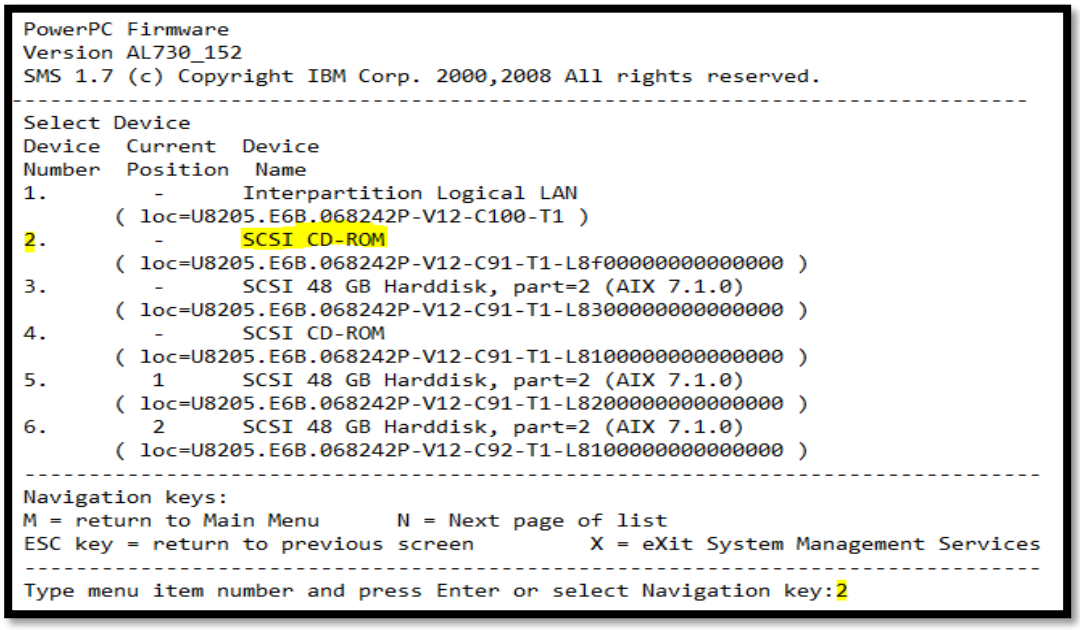
4.5 In Boot Options, Choose “Select Install/Boot Device” by Type “1” in navigation key field



4.6 In Select Device Type options, Choose “List all Devices” by Type “7” in navigation key field

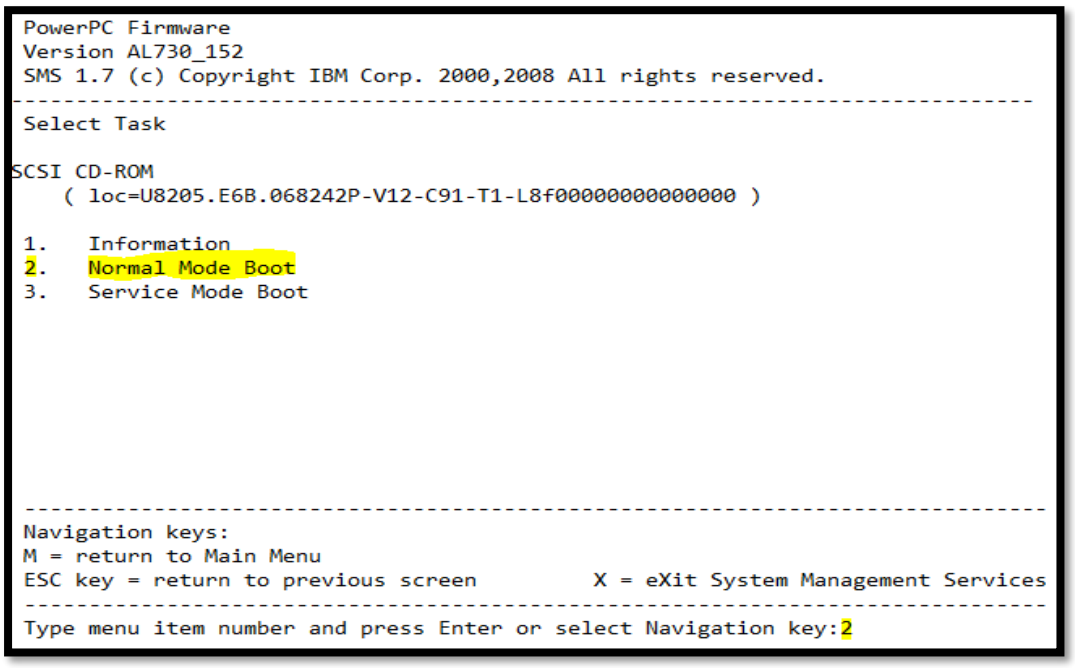


4.7 In Select Device menu, Choose CD-ROM by Type “2” in navigation key field

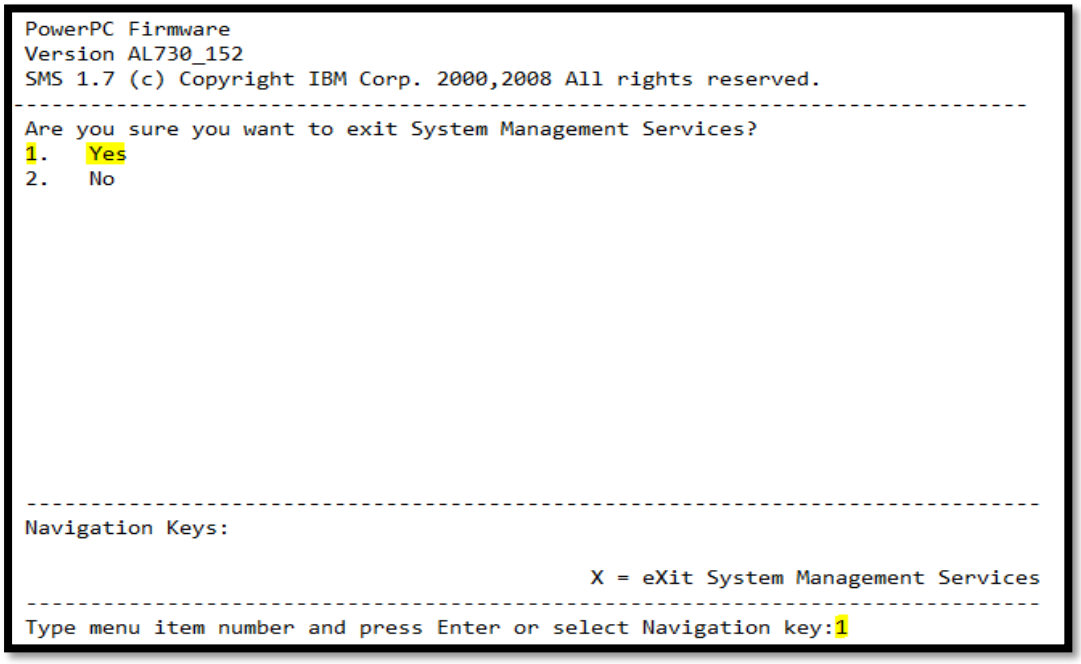


Note: SCSI CD-ROM is the virtual optical device mapped from VIO server

4.8 In Select Task Menu, Choose “Normal Mode Boot” by Type “2” in navigation key field

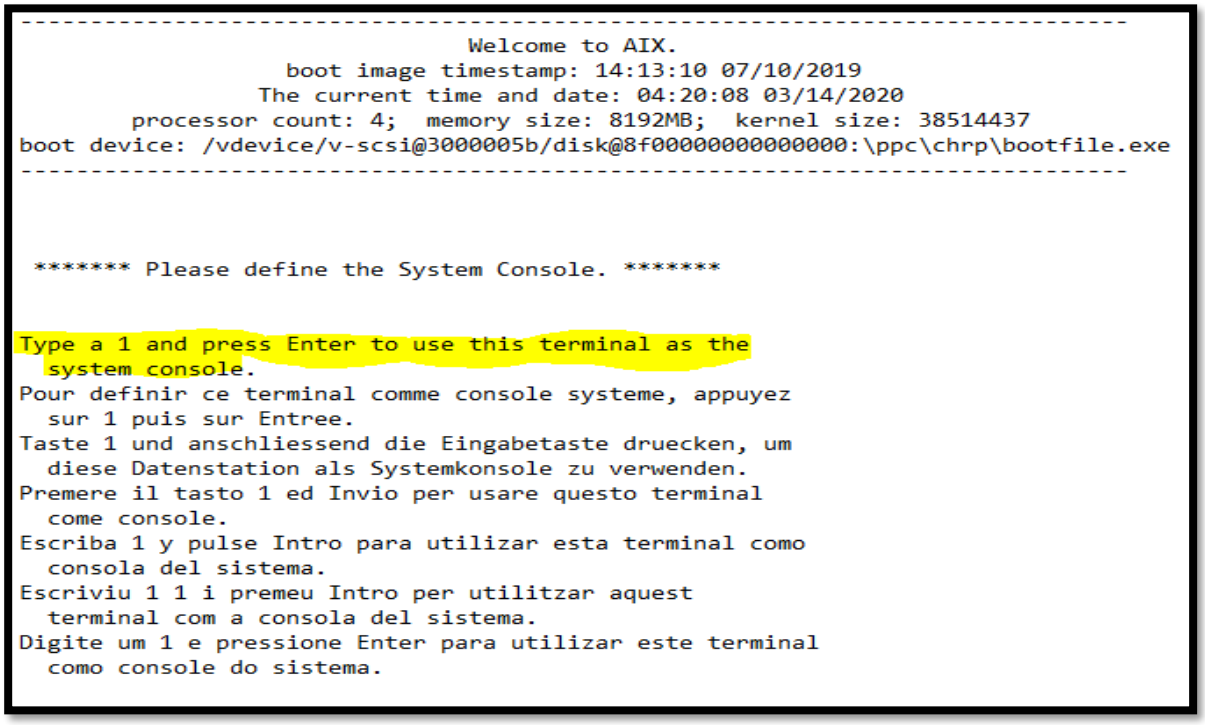


4.9 Save the Boot configuration and exit from SMS menu by Type “1” in navigation key field

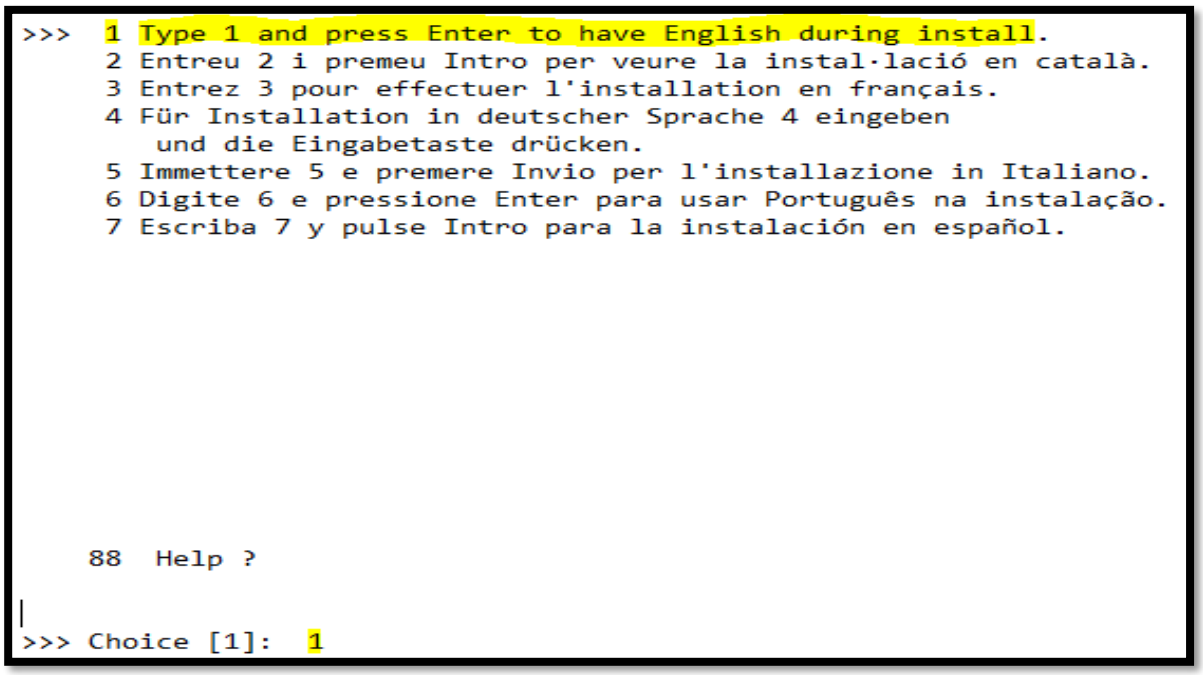


Note: Server will reboot after exit from SMS menu

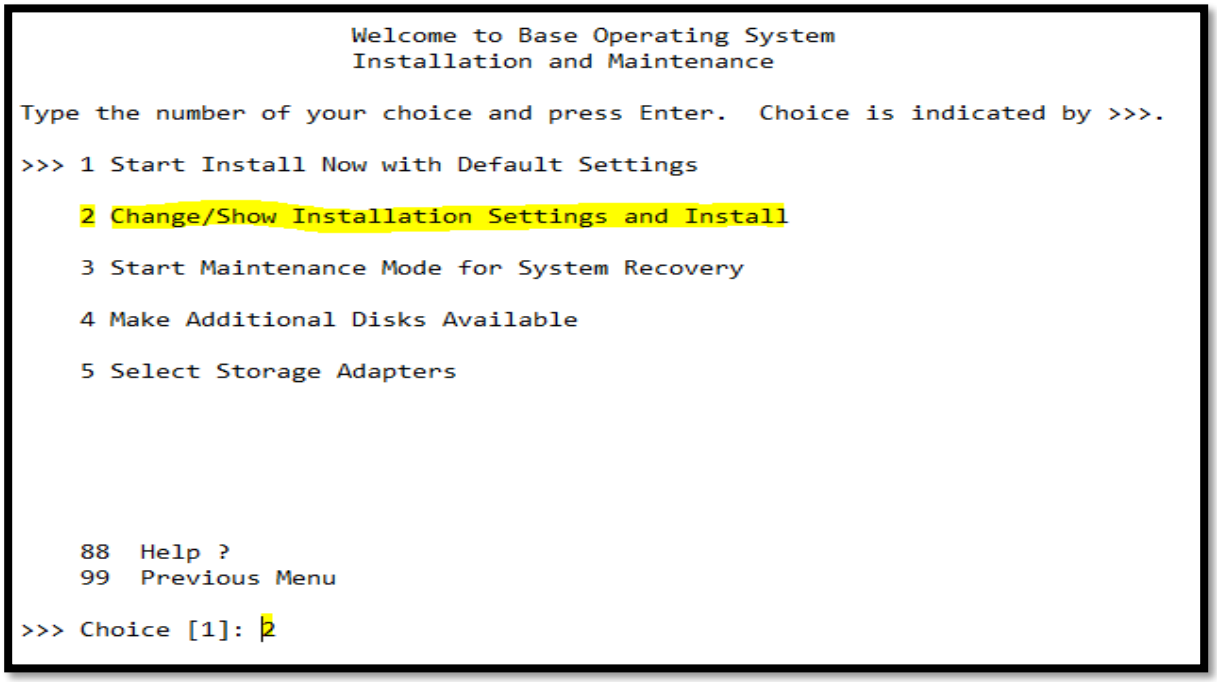
4.10 Now server will boot with CD-ROM which having Bootable BOS image. Now Type a 1 and press Enter to use this terminal as the system console



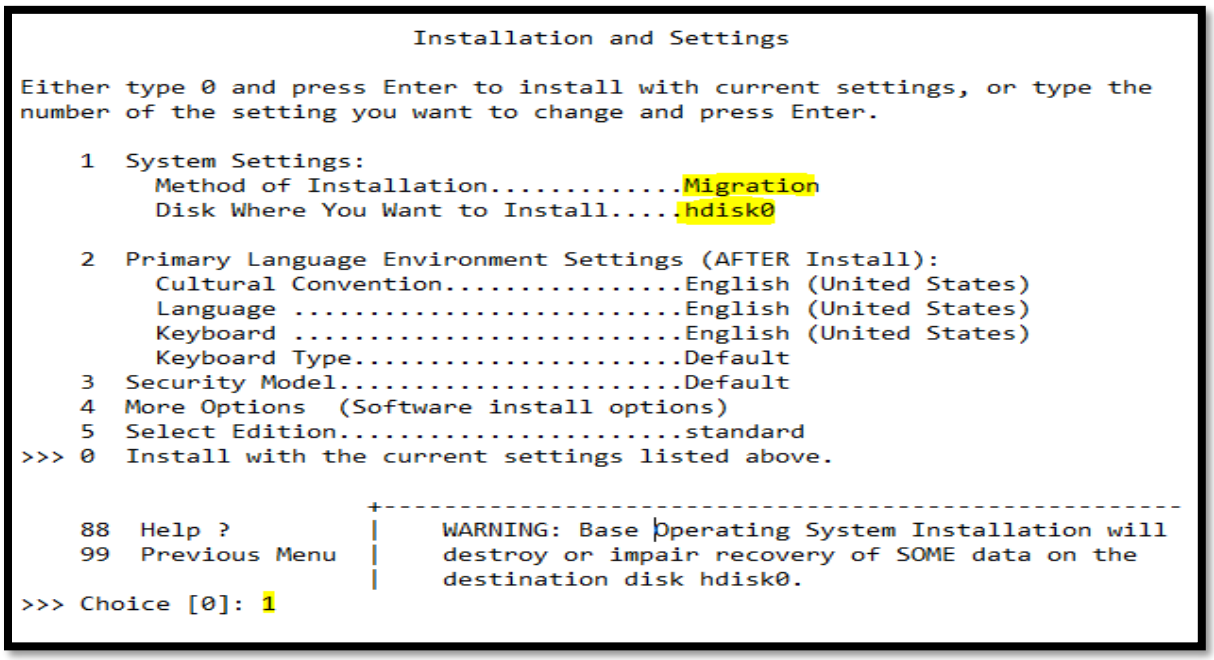
4.11 Select Installation Menu Language by Type 1 and press Enter to have English during install. Now BOS installation and Maintenance window will open



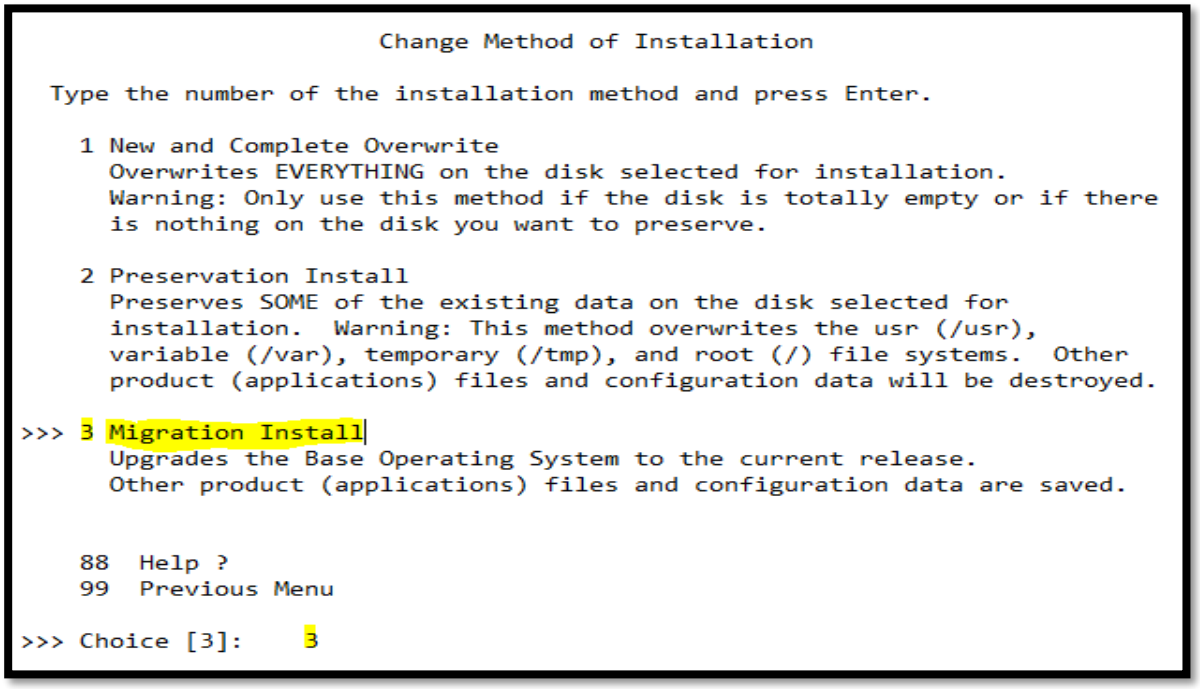
4.12 Type 2 and Choose “Change/Show Installation Settings and Install” in BOS Installation and Maintenance



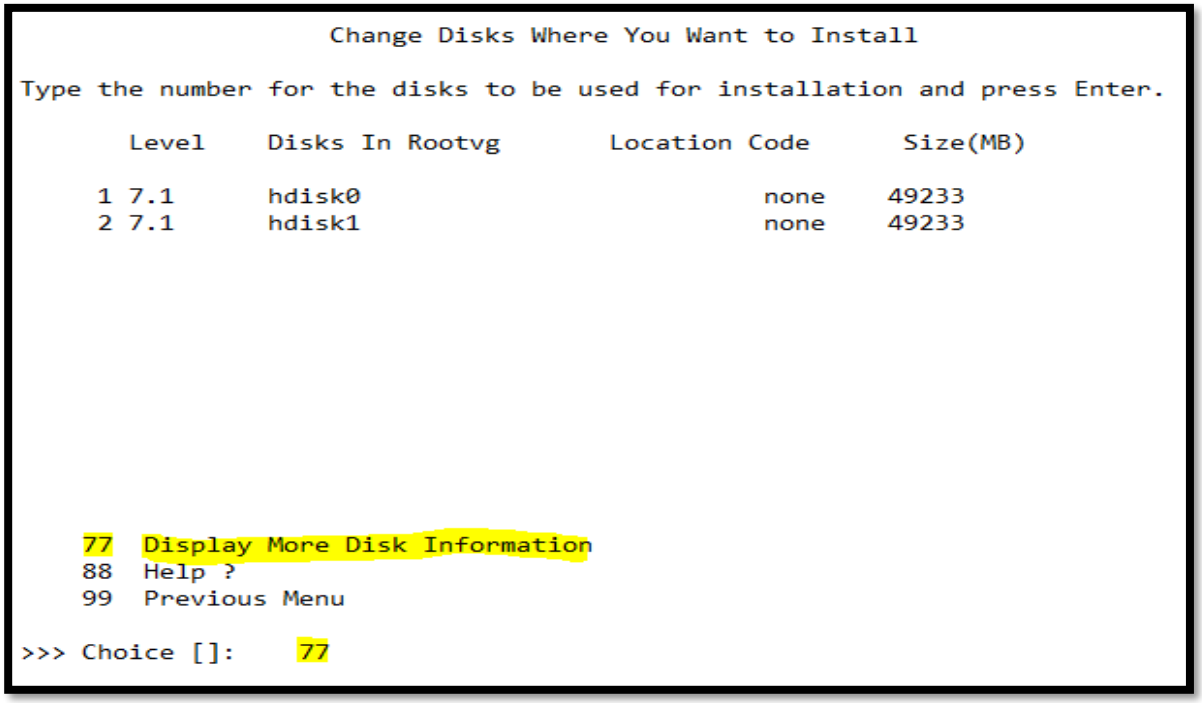
4.13 In the Installation and Settings window, Type 1 and press enter to select “System Settings”, then Change method of Installation menu will open.



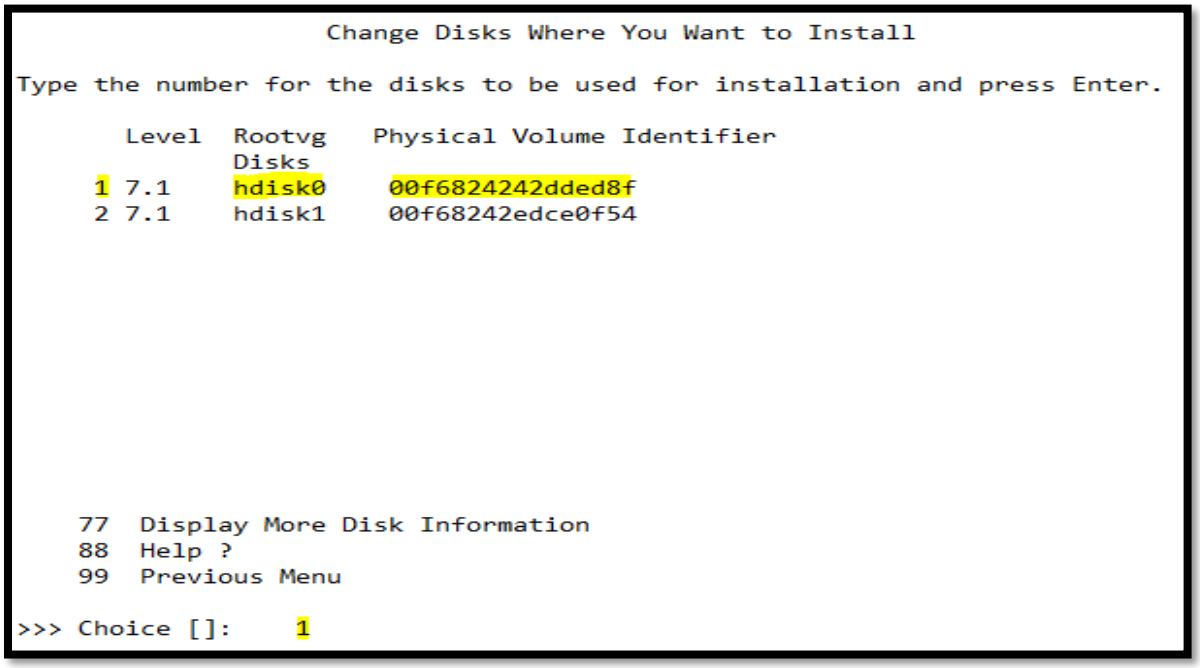
4.14 Change method of Installation menu, Type 3 and press enter to choose Migration Install. Now "Change Disks where You Wants to Install" menu will open



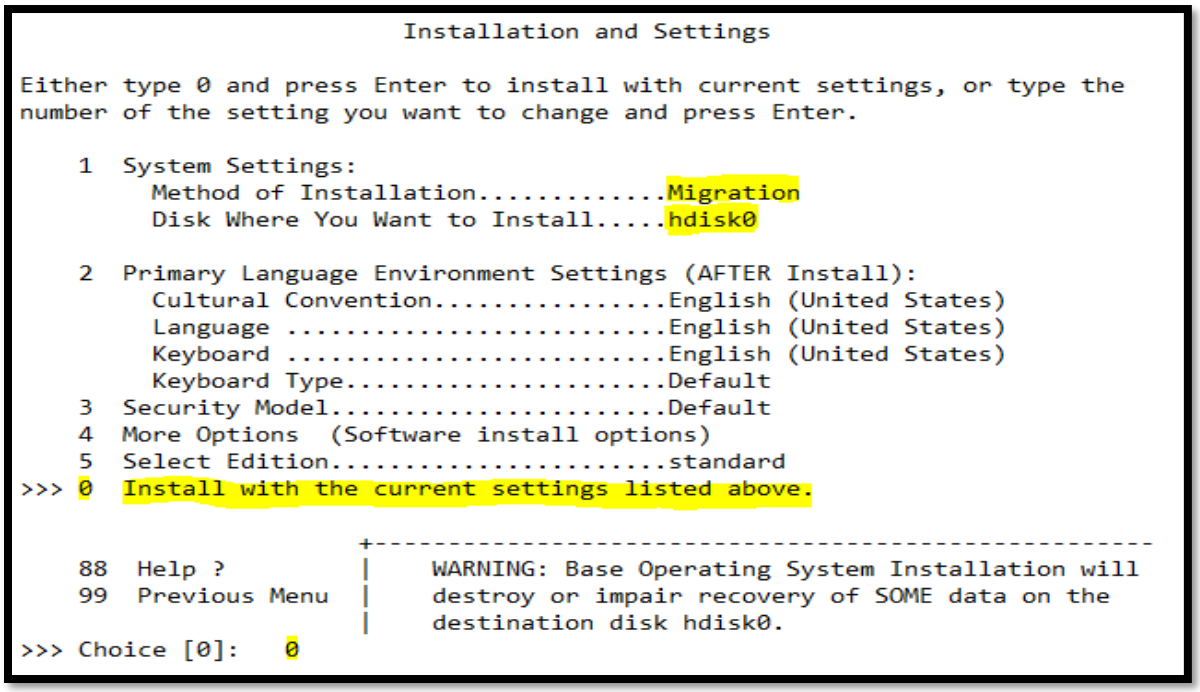
4.15 In Change Disks where You Wants to Install menu, Type "77" and press enter to Choose "Display More Disk Information"



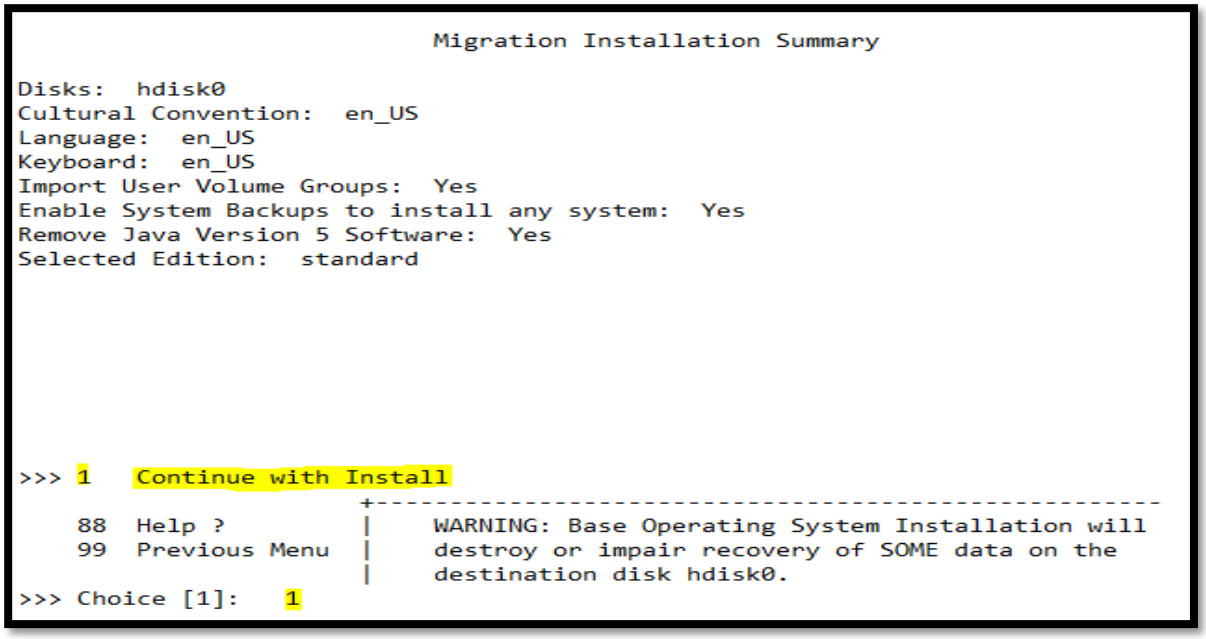
4.16 Identify the PVID of root vg disk, where we upgrade the OS. In this list Select option 1 So that you perform OS migration on hdisk0 and hdisk1 having old OS for roll back if migration fails



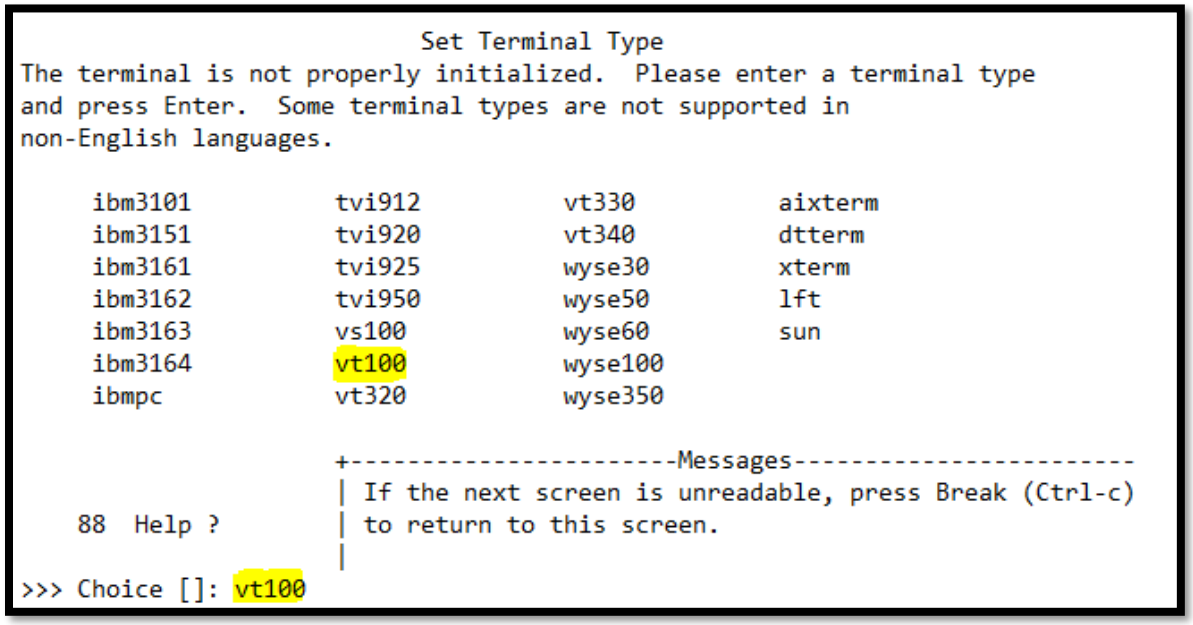
4.17 In the Installation and Settings window, Type “0” and press enter to select “Install with the current settings listed above”.



4.18 Review the summary and Type “1” and press enter to select “continue with Install”

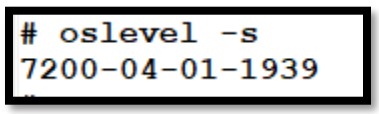


4.19 After completion file sets installation set the terminal type as **vt100** and press enter.



4.20 Once server came up. Check the OS level by issuing the below command.

oslevel -s



4.21 Check the known recommended maintenance levels by issuing below command.

oslevel -rq

```
# oslevel -rq
Known Recommended Maintenance Levels
------------------------------------
7200-04
7200-03
7200-02
7200-01
7200-00
```

4.22 Check the known service packs by issuing the below command

oslevel -sq

```
# oslevel -sq
Known Service Packs
-------------------
7200-04-01-1939
7200-04-01-1938
7200-04-00-0000
7200-03-04-1938
```

4.23 Once Application team confirm we can mirror the rootvg Disks by issuing the below command.

mirrorvg rootvg hdiskX (Here hdisk4 is the new free physical volume)

```
root@u0bnim001 /> mirrorvg rootvg hdisk4
root@u0bnim001 />
```

4.24 For rollback, change the bootlist and reboot the server, server will boot with Old OS version and it packages. Post we can Mirror the rootvg disks

Change the bootlist order by issuing the below command

bootlist -m normal hdisk3 (Here hdisk3 is old rootvg physical volume)

Check the bootlist order by issuing the below command.

bootlist -m normal -o

```
root@u0bnim001 /> bootlist -m normal hdisk3
root@u0bnim001 />
root@u0bnim001 /> bootlist -m normal -o
hdisk3 blv=hd5 pathid=0
hdisk3 blv=hd5 pathid=1
```

5. VERSION HISTORY

	Prepared By	Reviewed by	Approved By
Name	Kenche Jayakrishna	Franklin K Dhas	Arunachalam G
Role	Unix L2	SME	Senior Manager
Date	16-MAR-2020	17-MAR-2020	**27-MAR-2020**
Version	1.0	1.0	1.0